

Fig. 3: An actual-size, single-side PCB for the serial LCD module

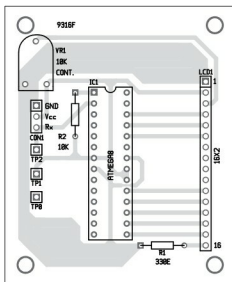


Fig. 4: Component layout for the PCB

main controller (Arduino Uno in this case). It acts as a bridge between the main controller and the LCD.

ATmega8 can be replaced by ATmega168/328 without changing the code, but the bootloader will need to be changed. Table 1 indicates the connections between IC1 and LCD1. The supply to the module is provided by the Arduino Uno itself as shown in Fig. 2.

**Arduino Uno board.** Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments. Arduino Uno is a board based on ATmega328 microcontroller. It comprises 14 digital input/output (I/O) pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

## Software

The software for the serial LCD module is written in Arduino programming

language. Follow the steps below to burn the bootloader and the firmware in IC1:

1. All the required software is provided in EFY DVD and at source. [efymag.com](http://efymag.com). Put the ATmega8 (IC1) chip in Arduino Uno board and follow the instructions at [http://playground.arduino.cc/Learning/Burn168.Ux17\\_noSX0](http://playground.arduino.cc/Learning/Burn168.Ux17_noSX0) to burn the bootloader (ATmega8 noxtal). The fuse bits are 0xCA and 0xD4. Program lock bits as 0x0F so that the bootloader is not overwritten every time you flash new firmware.

2. Edit board file (arduino-0022/hardware/arduino/boards.txt) by adding the lines:

```
atmega8noxtal.name=ATmega8-noxtal 8MHz
atmega8noxtal.upload.protocol=stk500
atmega8noxtal.upload.maximum_size=7168
atmega8noxtal.upload.speed=38400

atmega8noxtal.bootloader.low_fuses=0x1f
atmega8noxtal.bootloader.high_fuses=0xca
atmega8noxtal.bootloader.path=atmega8_noxtal
atmega8noxtal.bootloader.file=ATmega8BOOT.hex
atmega8noxtal.bootloader.unlock_bits=0x3F
atmega8noxtal.bootloader.lock_bits=0x0F

atmega8noxtal.build.mcu=atmega8
atmega8noxtal.build.f_cpu=800000L
atmega8noxtal.build.core=arduino
atmega8noxtal.build.variant=standard
```

3. Restart Arduino IDE (Arduino-0022) and you will see a new

## PARTS LIST

### Semiconductors:

- IC1 - ATmega8 microcontroller
- LCD1 - 16x2 LCD module

### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 330-ohm
- R2 - 10-kilo-ohm
- VR1 - 10-kilo-ohm preset

### Miscellaneous:

- BOARD1 - Arduino Uno board
- CON1 - 3-pin connector

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)

entry (ATmega8-noxtal@8MHz) in the 'Boards' menu. Copy the provided code in Arduino IDE and select board as ATmega8-noxtal.

4. Go to File→Upload to program the sketch in IC1.

5. Remove ATmega8 chip from the Uno board and place back the original microcontroller. Put the programmed ATmega8 chip in the provided PCB and your serial LCD module is ready to use. If you wish to use ATmega168/328, you must choose proper bootloader and board before uploading. Sketch is compatible with all these microcontrollers.

You can use serial LCD library if you want to drive the serial LCD module using Arduino Uno. For that, connect the board with the module as shown in Fig. 2 and copy the library (serialLCD) to the libraries folder of Arduino IDE.

You can use `lcd.backLight()` function to control the brightness of LCD.

## Construction and testing

An actual-size, single-side PCB for serial LCD module is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the recommended PCB to minimise any assembly errors. Use IC base for ATmega8.

To test the circuit for proper functioning, verify correct 5V supply at TP1 with respect to TP0. The serial data to the LCD module can be seen at TP2 using an oscilloscope. ●

The author is a third-year diploma student at Government Polytechnic, Kolhapur